

HENRY DOWD

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PROFILE

BSc in Pure Mathematics & Computer Science, Maynooth University, completed May 2026, with a year abroad at the University of Ottawa. Run a two-node k3s cluster on a single Proxmox host: hardware and virtualisation up through GitOps-managed Kubernetes, with CI/CD, monitoring, and alerting. Troubleshoot faults across the stack, from hardware quirks and kernel drivers up to Kubernetes workloads. Targeting platform engineering, site reliability, and low-level infrastructure.

TECHNICAL SKILLS

Linux & Virtualisation Arch Linux, Proxmox VE, LXC, ZFS, systemd, shell scripting
Networking & Security TCP/IP, DNS (split-horizon), WireGuard, Cloudflare Tunnel + Access, SSL/TLS
Kubernetes & GitOps k3s, Helm, ArgoCD (app-of-apps), MetalLB, Traefik, Longhorn, Sealed Secrets, cert-manager
CI/CD & Observability GitHub Actions, GHCR, VictoriaMetrics, Grafana, Alertmanager, restic, Backblaze B2
Languages & Tools Python, Bash, Java, C, SQL; Git, Gitea, Technitium DNS

PROJECTS

Homelab — k3s / GitOps Platform · Personal

2024 – Present

- Two-node k3s cluster on a single Proxmox host, GitOps via ArgoCD app-of-apps pattern, git repo as single source of truth.
- Platform assembled from components: MetalLB, Traefik ingress, Longhorn persistent block storage (replica=1, deliberate single-worker tradeoff over host-level ZFS RAID-1 mirror), Sealed Secrets for encrypted credentials in git.
- Cluster-wide hardening: NetworkPolicy default-deny per namespace, non-root SecurityContext with dropped capabilities and read-only root filesystem.
- CI/CD end-to-end: GitHub Actions builds, pushes to GHCR, updates SHA-pinned manifest, ArgoCD reconciles; in-cluster runner lints manifests.
- Public ingress via Cloudflare Tunnel (no inbound ports open), split-horizon DNS via Technitium, wildcard TLS via cert-manager and Let's Encrypt DNS-01; Cloudflare Access on sensitive services.
- Observability with VictoriaMetrics, Grafana, Alertmanager and `absent()` guards so a dead scrape self-alerts; encrypted offsite restic backups to Backblaze B2, quarterly verified restores.
- Diagnosed intermittent e1000e (Intel I219-LM) NIC transmit hangs across ethtool offload settings, modprobe driver parameters, and GRUB `pcie_aspm=off`.
- Architecture decisions recorded as ADRs, incidents as post-mortems. github.com/hpdowd/homelab

Final Year Project — Paraphrase Detection · Maynooth University

2025 – 2026

- Built a dual-branch classifier fusing BERT cosine similarity, Zhang–Shasha tree edit distance, Jaccard overlap, and length ratio; evaluated on the MSRP corpus in Python with HuggingFace Transformers, PyTorch, and scikit-learn.
- Performed error analysis comparing structural vs. semantic methods, with confusion matrices, score distributions, and parse-tree examples of failure modes. github.com/hpdowd/paraphrase_detector

PDF-Sorter — Workplace Automation Tool · Python

2025

- Regex-based tool to auto-sort and rename PDFs by filename and content metadata, with OCR fallback; deployed during HR placement at Roscommon County Council. github.com/hpdowd/PDF-Sorter

EXPERIENCE

Summer Student — Human Resources · Roscommon County Council

Jun – Sep 2025

- Data entry, records management, and financial processing across employee files and payroll administration.
- Identified a repetitive document-sorting bottleneck and shipped the PDF-Sorter tool to automate it.

EDUCATION & CERTIFICATIONS

BSc Computational Thinking · Maynooth University

Sep 2022 – May 2026

Pure Mathematics & Computer Science

Study Abroad · University of Ottawa

Sep 2024 – May 2025

Fortinet NSE 1–3 (2026) · Full driving licence · CTYI alumnus